Data Lakehouse with Snowflake

Big Data Engineering | 1st Assignment

SPRING 2025

STACK:



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Introduction

YouTube (the world-famous video sharing website) maintains a list of the top trending videos on the platform. According to Variety magazine, “To determine the year’s top-trending videos, YouTube uses a combination of factors including measuring users' interactions (e.g. number of views, shares, comments and likes).

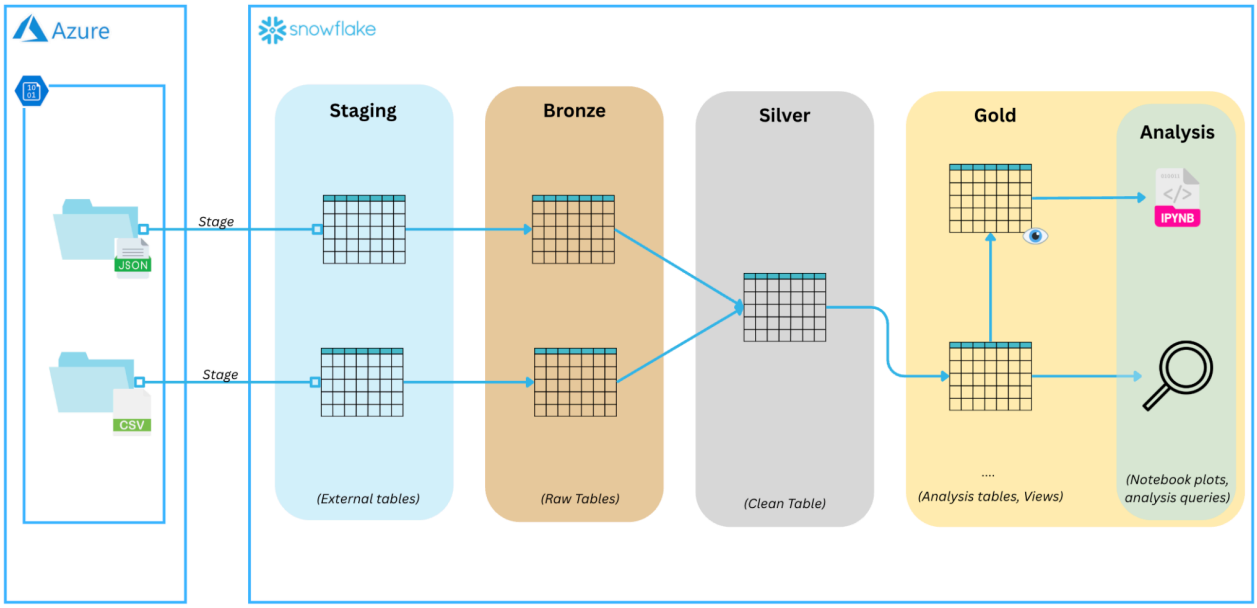
Dataset

A dataset with a daily record of the top trending YouTube videos has been extracted through the Youtube API and made available on the Kaggle.
(https://www.kaggle.com/rsrishav/youtube-trending-video-dataset)

This dataset includes several months (from 2020-08-12 to 2024-04-15) of data of daily trending YouTube videos. Data is included for the IN, US, GB, DE, CA, FR, BR, MX, KR, and JP regions (India, USA, Great Britain, Germany, Canada, France, Brazil, Mexico, South Korea, and Japan respectively), with up to 200 listed trending videos per day.

The data also includes a category_id field, which varies between regions. To retrieve the categories for a specific video, find it in the associated JSON. One such file is included for each of the 10 regions in the dataset.

High level Architecture

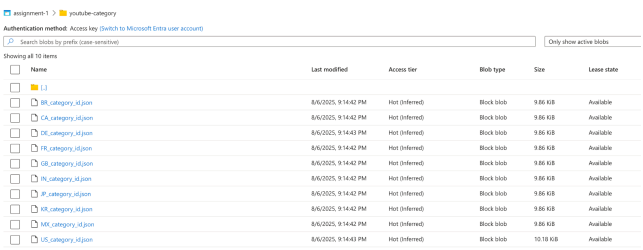


Overview of the project

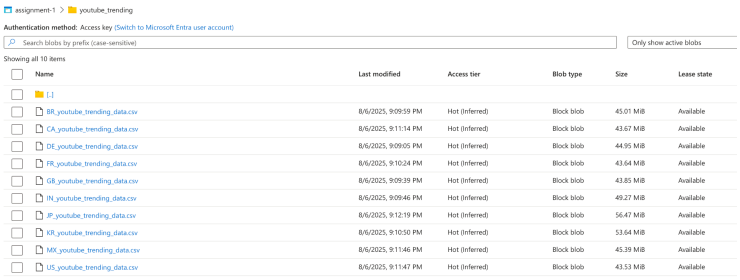
Staging

The process involves downloading source files from Kaggle and storing them in an Azure storage container named **assignment-1**. This container is organized into two directories:

- **youtube-category**: This directory holds JSON files that define the video categories for ten countries.



- **youtube_trending**: This directory contains CSV files with trending video statistics for each country, covering the period from August 12, 2020, to April 15, 2024.



To begin the data ingestion process within Snowflake, a database named **assignment_1** was created. Within this database, the data pipeline was configured by creating several key objects in sequence.

First, two stages were established to securely connect to the Azure storage container using a Shared Access Signature (SAS) token:

- **stage_assignment_youtube_trending** for the trending video CSV files.
- **stage_assignment_youtube_category** for the category definition JSON files.

Next, two specific file formats were defined to correctly interpret the structure of the source files:

- **file_format_csv** to handle the structure of the CSV files.
- **file_format_json** to parse the JSON data.

Finally, leveraging these stages and formats, two external tables were created within the database's default **public** schema:

- **ex_table_youtube_trending**, which points to the CSV data.
- **ex_table_youtube_category**, which points to the JSON data.

Bronze Layer

The next architectural step was to build the **Bronze layer**, which involved creating a new schema named **bronze** inside the **assignment_1** database. This schema is designed to hold the first physical, cleaned, and properly typed copy of the data stored within Snowflake.

Two tables were created in this bronze schema by selecting data from the external tables:

1. **Bronze.table_youtube_trending**: Was created to store the trending video statistics with the following structure:

Column Name	Data Type
VIDEO_ID	VARCHAR
TITLE	VARCHAR
PUBLISHEDAT	TIMESTAMP_NTZ
CHANNELID	VARCHAR
CHANNELTITLE	VARCHAR
CATEGORYID	NUMBER
TRENDING_DATE	DATE

VIEW_COUNT	NUMBER
LIKES	NUMBER
DISLIKES	NUMBER
COUNTRY	VARCHAR

2. **Bronze.table_youtube_category** table was created to hold the category definitions with this schema:

Column Name	**Data Type**
COUNTRY	VARCHAR
CATEGORY_ID	VARCHAR
CATEGORY_TITLE	VARCHAR

Silver Layer

In this layer, the data from the Bronze layer undergoes analysis, cleaning, and transformation. A new schema, **silver**, was created within the **assignment_1** database to house these enriched datasets.

The first step was to create a combined table called **silver.table_youtube_final**. This table was built by joining **bronze.table_youtube_trending** and **bronze.table_youtube_category** on the **country** and **categoryid** fields. A **LEFT JOIN** was used to ensure no records from the trending table were lost. A new unique identifier column, **ID**, was also added using the **UUID_STRING()** function.

The schema for this final table is as follows:

Column Name	Data Type
ID	VARCHAR(36)
VIDEO_ID	VARCHAR(16777216)
TITLE	VARCHAR(16777216)
PUBLISHEDAT	TIMESTAMP_NTZ(9)
CHANNELID	VARCHAR(16777216)
CHANNELTITLE	VARCHAR(16777216)
CATEGORYID	NUMBER(38,0)

TRENDING_DATE	DATE
VIEW_COUNT	NUMBER(38,0)
LIKES	NUMBER(38,0)
DISLIKES	NUMBER(38,0)
COUNTRY	VARCHAR(16777216)
CATEGORY_TITLE	VARCHAR(16777216)

During the analysis of this layer, several data quality checks were performed. The following questions were investigated and actions were taken:

1. ***In "table_youtube_category" which category_title has duplicates if we don't take into account the categoryid?***

Finding: The category title **'Comedy'** was identified as a duplicate.

2. ***In "table_youtube_category" which category_title only appears in one country?***

Finding: The category **'Nonprofits & Activism'** appears only in the **US** data.

3. ***In "table_youtube_final", what is the categoryid of the missing category_title?***

Finding: The **categoryid** for all rows with missing titles is **29**.

4. ***Update the table_youtube_final to replace the NULL values in category_title with the answer from the previous question.***

Action: The update was successfully performed, affecting **1,563 rows** with 'Nonprofits & Activism' as the new category title.

5. ***In "table_youtube_final", which video doesn't have a channeltitle (return only the title)?***

Finding: The video title is **"Kala Official Teaser | Tovino Thomas | Rohith V S | Juvis Productions | Adventure Company"**.

6. ***Delete from "table_youtube_final", any record with video_id = "#NAME?"***

Action: All records with this invalid **video_id** were deleted, resulting in the removal of **32,081 rows**.

7. ***Create a new table called "table_youtube_duplicates" containing only the "bad" duplicates by using the row_number() function.***

Action: A new table was created in the silver schema to isolate the duplicate rows for removal.

```
-- Creates a new table containing only the duplicate records from the final table.
CREATE OR REPLACE TABLE silver.table_youtube_duplicates AS
SELECT * EXCLUDE(test_row)
FROM (
    SELECT
    *,
    ROW_NUMBER() OVER (
        PARTITION BY video_id, country, trending_date
        ORDER BY view_count DESC
    ) as test_row
    FROM silver.table_youtube_final
    ORDER BY 1,2,3
)
WHERE test_row != 1;
```

8. ***Delete the duplicates in "table_youtube_final" by using "table_youtube_duplicates".***

Action: The identified duplicates were then deleted from the main table by matching their unique IDs. **37466 rows.**

```
-- Deletes all duplicate records from the final table by matching them with the duplicates table.
DELETE FROM silver.table_youtube_final yf
    WHERE EXISTS(
    SELECT 1
    FROM silver.table_youtube_duplicates yd
    WHERE yf.id = yd.id
);
```

9. ***Count the number of rows in "table_youtube_final" and check that it is equal to 2,597,494 rows.***

Action: This query will return TRUE if the count is correct.

```
SELECT COUNT(*) = 2597494 FROM silver.table_youtube_final;
```

Gold Layer

The Gold Layer is the final stage of the pipeline, focused on business intelligence and analytics. In this part, the clean, transformed data from the Silver Layer is used to generate insights and answer specific business questions, achieved by creating aggregated views that are optimized for visualization and reporting in tools like data notebooks.

the analysis in the Gold Layer was expected to answer the following questions:

1. What are the top 3 most viewed videos for each country in the 'Gaming' category for the trending date of 2024-04-01?

e.g.

COUNTRY	TITLE	CHANNELTITLE	VIEW_COUNT	RK
BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
BR	IShowSpeed x MC Kevin O Chris - Amar de (Official Music Video)	IShowSpeed	2971782	2
BR	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3

CA	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
CA	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
CA	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3
DE	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
DE	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
DE	Season 3 Warzone Launch Trailer - Rebirth Island | Call of Duty: Warzone	Call of Duty	2311131	3

2. For each country, what is the total count of distinct videos with a title containing the word “BTS” (case-insensitive)?

COUNTRY	CT
KR	468
IN	288
US	268
CA	262
MX	254

JP	251
DE	242
GB	223
BR	186
FR	167

3. For each country and for each month before the year 2024, what was the single most-viewed video?

e.g.

COUNTRY	YEAR_MONTH	TITLE	CHANNELTITLE	CATEGORY_TITLE	VIEW_COUNT	LIKES_RATIO
BR	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	139504939	3.20
CA	2024-01-01	Still Here | Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69

DE	2024-01-01	Still Here | Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69
FR	2024-01-01	Still Here | Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69
GB	2024-01-01	Still Here | Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69
IN	2024-01-01	Protect $500,000 Keep It!	MrBeast	Entertainment	85458562	4.21
JP	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	137639799	3.22

KR	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	143955997	3.16
MX	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	137639799	3.22

4. For each country and for all data before 2022, which category_title had the most distinct videos?

e.g.

COUNTRY	CATEGORY_TITLE	TOTAL_CATEGORY_VIDEO	TOTAL_COUNTRY_VIDEO	PERCENTAGE
BR	Entertainment	5417	23760	22.80
DE	Entertainment	7709	30719	25.10
FR	Entertainment	7548	32849	22.98
GB	Entertainment	5643	27855	20.26
IN	Entertainment	21281	50250	42.35
JP	Entertainment	5658	17627	32.10
KR	Entertainment	5122	15175	33.75
MX	Entertainment	4195	17532	23.93
CA	Gaming	6594	30869	21.36

US	Gaming	6226	28799	21.62

5. Which channeltitle has produced the highest number of distinct videos overall, and what is this number?

CHANNELTITLE	TOTAL_COUNT
Vijay Television	2049

6. If you were to launch a new Youtube channel tomorrow, which category (excluding “Music” and “Entertainment”) of video will you be trying to create to have them appear in the top trend? Will this strategy work in every country?

To formulate a robust launch strategy for a new YouTube channel, the analysis focuses on the period **from 2022 onwards**. This timeframe is critical because it allows to examine the most **current digital trends** and evaluate their potential to **persist over time**. This analysis follows a two-tiered approach. It begins at a **global level** to identify broad viewership patterns and form an initial hypothesis about successful content. Subsequently, this hypothesis is validated through a **country-specific analysis** to ensure that these global insights are relevant and actionable within a targeted local market.

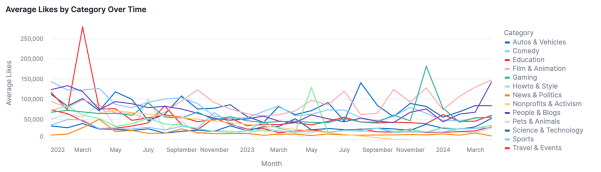
The first step is to understand which categories are performing well on a worldwide scale:

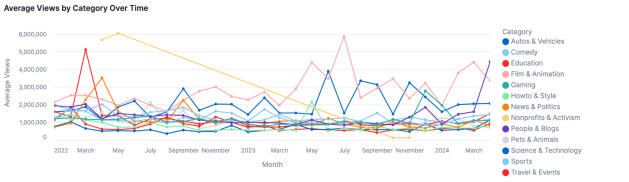
- **Table Created:** `gold.table_youtube_analysis_global`
  - **Purpose:** This table acts as the foundation for the global strategy. It aggregates data across all countries to calculate important metrics like the number of unique channels (a proxy for **competition**), average views and likes (a proxy for **popularity**), and the average engagement ratio (a proxy for **audience connection**).
- **Views Created:** `gold.v_yt_global_*` (e.g., `v_yt_global_unique_channels`, `v_yt_global_avg_engagement_ratio`, etc.)

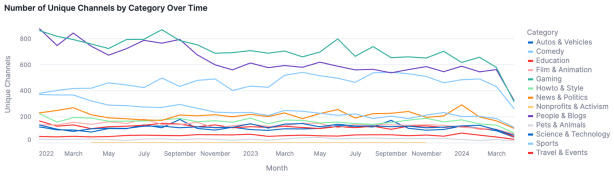
The second step is to determine if the promising global strategy is actually viable in each target country:

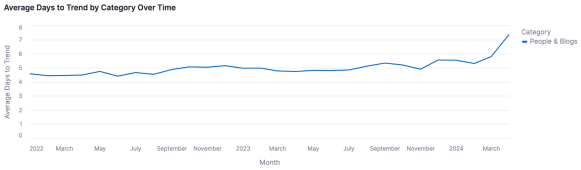
- **Table Created:** `gold.table_youtube_analysis_source_by_country`
  - **Purpose:** This table breaks down the global data by country.
- **Views Created:** `gold.v_yt_country_*` (e.g., `v_yt_country_avg_views`, `v_yt_country_total_likes`, etc.)

Opportunities and Limitations









The global analysis identifies **'People & Blogs' as the strongest candidate** for achieving top-200 trending status. This conclusion is driven by its consistently rising trend in average views and likes since early 2024. While it is the most competitive category and now takes nearly eight days for a video to trend, its worldwide dominance makes it the premier choice for a broad, global strategy.





However, a deeper dive into regional data reveals that a one-size-fits-all approach isn't optimal. Strong alternative categories emerge as powerful contenders in specific markets. For example, **'Science & Technology' is a top performer in South Korea and Mexico.**

This highlights a crucial strategic choice:

- For a **global campaign** aiming for the widest possible reach, **'People & Blogs' remains the most viable path**, despite its high competition.
- For a more **targeted, country-specific strategy**, leveraging regional preferences by focusing on categories like 'Film & Animation' or 'Science & Technology' would be a more effective and efficient approach. This tailored plan allows for capitalizing on demonstrated local tastes rather than defaulting to the globally popular option.